The Role of Emojis in Sentiment Analysis of Financial Microblogs

Ahmed Mahrous
College of Finance and Economics
Qatar University
Doha, Qatar
0000-0003-4694-5336

Jens Schneider
College of Science and Engineering
Hamad Bin Khalifa University
Qatar Foundation, Doha, Qatar
0000-0002-0546-2816

Roberto Di Pietro, *Fellow, IEEE*
RC3, CEMSE Division
KAUST
Thuwal, Saudi Arabia
0000-0003-1909-0336

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***Abstract*—The application of sentiment analysis to the financial sector is a field that has been revamped thanks to social media, which has unleashed a trove of data to analyze. In particular, text analysis techniques have benefited from the attention that large part of data science researchers have devoted to it. However, as demographics evolve, so do the communication forms on social media. In particular, the usage of emojis to carry whole concepts is more and more diffused, though research on the topic is lacking. That is exactly the gap that we intend to cover with this contribution. In particular, after collecting more than 18.5 million posts from StockTwits, we use different supervised learning models in order to determine the role of emojis in sentiment analysis of financial posts on social media. We assess model accuracy, training/prediction speed, and sensitivity to training data set size for both emojis-only and text-only data, using logistic regression and BiLSTM models.**

Our main findings are staggering; we are the first to show that, when training sentiment analysis models exclusively on emojis, compared to a text-only approach: (i) achieved accuracy is competitive; (ii) training is 32 times faster; (iii) prediction times are reduced to a third; and, (iv) 40 times less data is needed to train the model. Additionally, we show some interesting patterns regarding emoji usage in financial microblogs. The cited contributions, other than being interesting on their own, also pave the way for further research in the field.

***Index Terms*—sentiment analysis, emojis, finance, microblogging, social media, machine learning, LSTM**

I. Introduction

More than 4.8 billion people use social media around the world, with an average usage of more than 2 hours per day [1]. Needless to say, social media is a significant technology that has become a major way for people to communicate. Social media can be used to connect with others, share ideas, and stay informed about current events and news. Social media also serve as a valuable communication tool for financial market participants, offering them a way to express their opinions, stay up-to-date with financial news, and gain insights from the perspectives of others.

Analyzing the sentiment of social media posts related to financial markets has significant value. For instance, the sentiment of financial posts can be used to construct an index of general investor sentiment [2]. Social media sentiment can also be used to build potentially profitable investing strategies. There are even investment funds focusing solely on sentiment analysis [3]. Sentiment analysis can also offer valuable information for researchers seeking to understand financial market behavior [4]. Sentiment analysis for financial markets can be different from sentiment analysis in other contexts. Some words carry meanings specific to financial markets (e.g.: bull/bear). Other words regarded as stop words in general sentiment become especially important in a financial context (e.g.: up/down). To have an idea of the subtleties related to this type of analysis, it is worth noting that sentiment analysis is required to interpret different semantics even within the same financial domain, for instance when analyzing formal financial texts, such as annual company reports, or financial microblogging text. Financial microblogging typically uses more informal language or slang, which would be rarely used in a more formal context. Financial microblogging can also have elements of speech that would very rarely be used in a more formal context, such as emojis.

Research questions. Pre-existing work has focused only on *adding* emojis to text to boost performance for sentiment analysis. In contrast, this paper seeks to answer the following research questions.

- Q1 How well can emojis alone serve as a proxy for public sentiment in finance and is there a performance degradation vs. using the textual part of a social media post?
- Q2 How much data is needed to successfully train logistic regression and bidirectional LSTM-based models to predict financial trends ("bullish", "bearish", "neutral") in social media micro-messages?

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More specifically, Q2 can be seen as a measure of the correlation between the components of messages (in our case, text and emojis in StockTwits) and the implied financial sentiment. Given that the narrow use of emojis in the financial space is not as heavily subjected to cultural biases as general-purpose use [5], [6], Q1 also implies an investigation if, in this context, emojis transcend languages.

Contributions. This paper sheds light on the added value provided by emojis in the context of sentiment analysis for financial markets—with the analysis conducted on data provided by microblogging platforms. In particular, we contribute a structured analysis of the sentiment prediction performance obtained using only emojis vs. only text—pre-existing work has focused only on *adding* emojis to text to boost performance. Our work is supported by a large amount of data (18.5 million posts) combined with state-of-the-art analysis techniques —logistic regression and a bidirectional [7] Long Short-Term Memory (LSTM)-based model citeHochreiter:1997:LSTM. Results are staggering: in the context of finance, we are the first to show that emojis are indeed strong proxies for financial sentiment. We also show that using only emojis requires less data to train on, making them an attractive choice if models for the original language of a social media post are immature or unavailable.

II. Background

In this section, we provide the required info for the comprehension of the remainder of the paper.

Emojis. The word emoji is made up of two Japanese words: e (絵, "picture") + moji (文字, "letter"). Emojis are widely used by social media users and have also been utilized by researchers for sentiment analysis. Emojis are believed to be first developed by the Japanese company Softbank in 1997 [8]. However, they were popularized in the 2010s after being standardized and released on several platforms, including Apple's iOS 5 [9]. In 2015, Instagram engineers estimated that around 50% of the comments on Instagram included an emoji [10]. In 2021, it was estimated that more than 20% of Tweets have emojis [11].

Each emoji is assigned a unique code in the Unicode encoding system, similar to characters such as letters. For instance, the letter "A" is assigned the Unicode code point U+0041, whereas the emoji 🚀 is assigned U+1F680. Whereas this, in theory, allows them to be displayed consistently across different devices and platforms, in practice, their appearance may vary due to font, software, and device differences.

People can use emojis to emphasize text, substitute text or contradict text (see also Table I). Emojis can offer valuable information about a sentiment of a social media post. Emojis can also be useful for understanding the sentiment of posts in different languages. For a more in-depth review of emojis including individual, cultural, and platform use diversity ("biases") we refer the reader to Bai et al. [12].

TABLE I: Different ways to use emojis.

Usage	Example
emphasizing text	What a great market! 🚀
substituting text	What a 🚀 market!
contradicting text	What a great market! 💩 🩸 📉

StockTwits platform. StockTwits started in 2008 and has been called the Twitter of financial markets. It is a microblogging platform that allows users to post and share ideas about financial assets including stocks and cryptocurrencies. Reuters reported in 2022 that the platform has more than 6 million registered users [13] and in 2019, Stocktwits reported that more than 200,000 posts were shared per day [14].

StockTwits is a highly valuable data source for finance and computer science researchers. Firstly, virtually all messages posted on StockTwits are related to financial markets. Secondly, users on StockTwits have the option to label their posts either as bullish (positive) or bearish (negative). These features allow researchers to create large data sets of labeled posts related to financial markets. These data sets can be useful for several analysis tasks, including sentiment analysis using supervised learning algorithms. Fig. 1 shows a visual breakdown of the demographics of StockTwits users as reported by the web analytics company Similarweb [15].

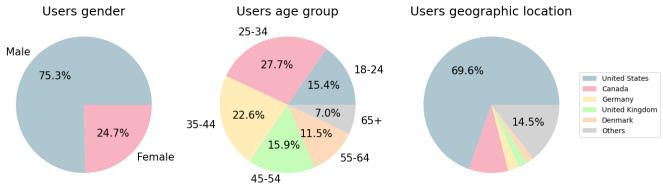

Fig. 1: Breakdown of StockTwits user demographics.

The majority of users on StockTwits are male. The age distribution is relatively dispersed, with the largest group being adults aged 25–44. Users are predominantly from western countries, especially the USA; this is reflected in the vast majority of posts being in English. Demography is relevant, since people of different genders, different ages and different cultures have been shown to use different informal words and emojis on social media [5], [6], [16].

III. Related Work

Given the role that emojis play in modern communications, they have attracted special attention from researchers in different fields, including marketing, psychology, medicine, education, and computer science [12]. Emojis have been used in sentiment analysis for social media posts of different domains: posts including consumer opinions of brands [17] or products [18], movie reviews [19],

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political posts [20], financial posts [21], and general domain posts [22].

An especially interesting aspect of emojis is their potential to transcending languages. A study found that emoji sentiment scores are similar for 13 different European languages [22]. Another study shows that sentiment scores are similar for many but not all emojis when comparing posts in English and Japanese [23]. Emojis have also been used in sentiment analysis of social media posts in Turkish [24], Chinese [25]–[28], Arabic [29]–[32] and several other languages [33], [34]. Emojis can also be used to detect more nuanced sentiments such as sarcasm, humor, or irony [28], [35], [36].

In the context of sentiment analysis, emojis have been used in different ways. Emojis have been used in the labeling process posts during pre-training, in order to overcome the unlabeled nature of social media posts. Prior work has assigned sentiment to Tweets based on the emojis they contain [37], and even has assumed that posts with emojis are non-neutral [28]. In addition, emojis have been used as inputs to sentiment analysis algorithms, considering emojis as special words [38]. In this context, a machine learning algorithm to convert each emoji into a dense vector representation to model its semantic meaning has been used. These emoji vector representations were then used as inputs to sentiment analysis algorithms to predict the sentiment of a certain post containing an emoji [38]. Emojis have also been used as outputs of sentiment analysis algorithms, where the sentiment of text can be analyzed to predict occurrences of emojis; this can be especially useful for emoji recommendation systems. Recurrent neural networks have also been used to predict emojis from text typed on mobile phone keyboards [39]. Some studies also assign sentiment scores to emojis. Schwartz et al. [40] first manually label 1.6 million Tweets as either positive, negative or neutral; the sentiment scores of emojis are then calculated based on the occurrences of each emoji in the Tweets and the labels assigned to those Tweets.

Prior research has demonstrated that adding emojis as inputs to sentiment analysis algorithms is useful [12], [21], [26], [27], [32], [38], [41]–[47], not only in the context of financial sentiment. One reason is that emojis often align with the meaning of the text and underscore its meaning. Therefore, emojis provide important clues to sentiment analysers.

One sub-domain where sentiment analysis and emojis can be of special interest is the domain of financial markets. Investor sentiment is a topic of significant interest in the finance field. There are several methods used to measure investor sentiment. The American Association of Individual Investors (AAII) Survey is a weekly survey, conducted since 1987, asking different groups of individual investors whether they think the direction of the stock market will be up (bullish), down (bearish), or experience no change (neutral) [48]. The survey results have been used as a measure of individual investor sentiment by many researchers in the field of finance [49], [50]. In addition to surveys, other proxy measures include the analysis of fund flows (e.g., to mutual funds), retail trading volumes, the premium paid for safe stocks, etc. [51].

A more recent trend is to measure investor sentiment by analyzing social media since it offers a relatively inexpensive way to directly measure investor sentiment. Dictionary-based sentiment analysis correlates with SP500 index intraday prices [52] and bitcoin prices [53]. However, later studies highlight the difference between financial social media posts and general-purpose ones, demonstrated by an improved performance of field-specific dictionaries [54].

Using machine learning for sentiment analysis, prior art also finds that the sentiment of online forum posts is related to future stock price movements [55], [56]. Simpler approaches show that even inspecting Tweets for keywords such as “bullish” and “bearish” has predictive value for stock prices [57].

Underscoring the importance of sentiment, Bloomberg terminal—a software platform that is widely used by professional financial analysts—offers social media sentiment indicators for stocks. Prior art shows a clear relationship between Bloomberg sentiment indicators and stock prices [58]–[60] but whether that is (considering the significance of the platform) a “self-fulfilling prophecy” is hard to analyze.

Some of the most recent work indicates that adding emojis as a non-textual form of data to the sentiment prediction process improves accuracy significantly, irrespective of whether a Naïve Bayes [21] or a deep neural network [41] is used.

In contrast to the prior art, which all add emojis to pre-existing text-based sentiment analysis, we investigate how much predictive power the *sole* use of emojis in financial social media posts carry.

IV. Data and Stylized Facts

We gathered over 18.5 million posts on StockTwits. Each of one these posts had the cashtag of at least one stock or cryptocurrency, as listed in Tab. II. The “Others” column in Tab. II represents a list of 14 stocks and cryptocurrencies including a gold ETF, AMC stock, and twelve cryptocurrencies.

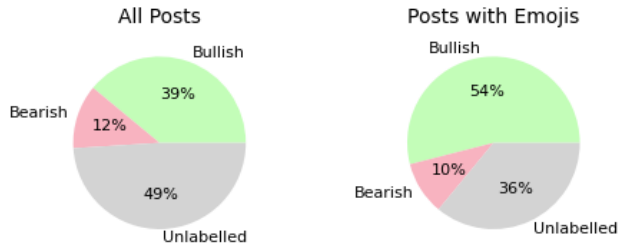

Fig. 2: User-assigned label breakdown in the data set.

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TABLE II: Number of posts by stock/cryptocurrency in the data sample.

Symbol	S&P 500	Bitcoin	Dogecoin	Apple	GameStop	Shiba Inu	Others
Number of Posts	6,067,248	2,803,824	2,236,581	1,897,368	1,386,519	1,220,177	2,945,419
Percentage	33%	15%	12%	10%	7%	7%	16%

Fig. 3: Left to right: Emoji clouds of posts labeled as bullish, bearish, or left unlabeled by poster.

Fig. 2 shows the proportion of posts based on label for all posts, and for posts that contain at least one emoji. Of the 18.5 million posts, approximately 39% are labeled as bullish, 12% are labeled as bearish and 49% are left unlabeled by the poster. Note that the fact that the user left the optional label unfilled does not mean that the post was meant to be neutral.

We only included labeled Tweets, of which 77% are labeled as bullish and 23% labeled as bearish. Around 14% of all posts include emojis. For posts that contain emojis, 54% are labeled as bullish, 10% as bearish and 36% are left unlabeled. Posts that contain emojis are, thus, more likely to be bullish.

The green "emoji cloud" diagram in Fig. 3 shows the emojis present in the posts labeled bullish by the poster, with the emoji size reflecting the frequency of the emoji. The larger the emoji, the higher the number of Tweets in which it is mentioned. Many of the larger emojis in the diagram clearly show a bullish sentiment such as 🔥 (fire, symbolizing that the market is "heating up"), 💎 (gem-stone, symbolizing that an asset is valuable), 🚀 (rocket, symbolizing that the market is going up), 🤑 (money-mouth-face, symbolizing that the poster is excited about making money), and 📈 (chart with upwards trend). Many of these emojis are much less used in non-financial contexts.

The red "emoji cloud" in Fig. 3 shows the emojis used in Tweets labeled as bearish by the poster. Some of the prominent emojis that most likely carry a bearish sentiment include 🐻 (bear-face), 🩸 (drop of blood, symbolizing losing or "bleeding"), 📉 (chart with downwards trend), 🚨 (police car revolving light, symbolizing an emergency or a crisis), and 💩 (pile of poo). Note that the posts labeled as bearish by users include some clearly bullish emojis, such as 🤑 or 📈, albeit at a lower frequency compared to the posts labeled as bullish.

The grey "emoji cloud" in Fig. 3 represents the emojis used in posts that are left unlabeled by the poster. Unlabeled posts are expected to be more bullish than bearish. This is also evident in the presence of more bullish emojis at a higher frequency in the "emoji cloud". However, some emojis seem to be larger than both the bullish and the bearish "emoji cloud", such as 🤷 (shrug emoji), 😆 (grinning squinting face) and 🙄 (face with rolling eyes). Note that some of the emojis present in the emoji clouds are modifiers such as 🏿 (skin-tone modifier) or ♂ (gender modifier). Emoji modifiers are added on top of original emojis in order to change certain characteristics such as the skin tone or gender of the emoji. Some emojis, such as 😂, are popular among internet users in general and not just in a financial context [61].

V. Methodology

In the following, we detail the adopted methodology that led to our results (see also Fig. 4 for a visual summary).

A. Pre-processing Steps

We perform the following pre-processing steps.

1. Balancing: The original data is highly unbalanced, with the majority of posts labeled by their poster as bullish instead of bearish. Training a machine learning model on an unbalanced data set can lead to results biased towards the majority class. For instance, if the data contains 77% of observations with a certain label (as in our case), the model can achieve 77% accuracy by simply predicting this label for all observations. To overcome this bias, we balance the data set by under-sampling the majority class: we take a random sample of posts from the bullish posts which are equal in number to the bearish posts.

2. Fix mojibake: Mojibake is the "garbled text that is the result of text being decoded using an unintended character encoding" [62]. We noticed that our data contained many instances of mojibake such as *"'", "â € " >", "â € ™ ","â € "* , etc. This mojibake has to be converted back to symbols before proceeding (see Sec. VI for details).

3. Lowercase all text: Lowercasing text before performing sentiment analysis is important because it allows for the standardization of input. For instance, without lowercasing the text, "Buy" and "buy" would be treated like

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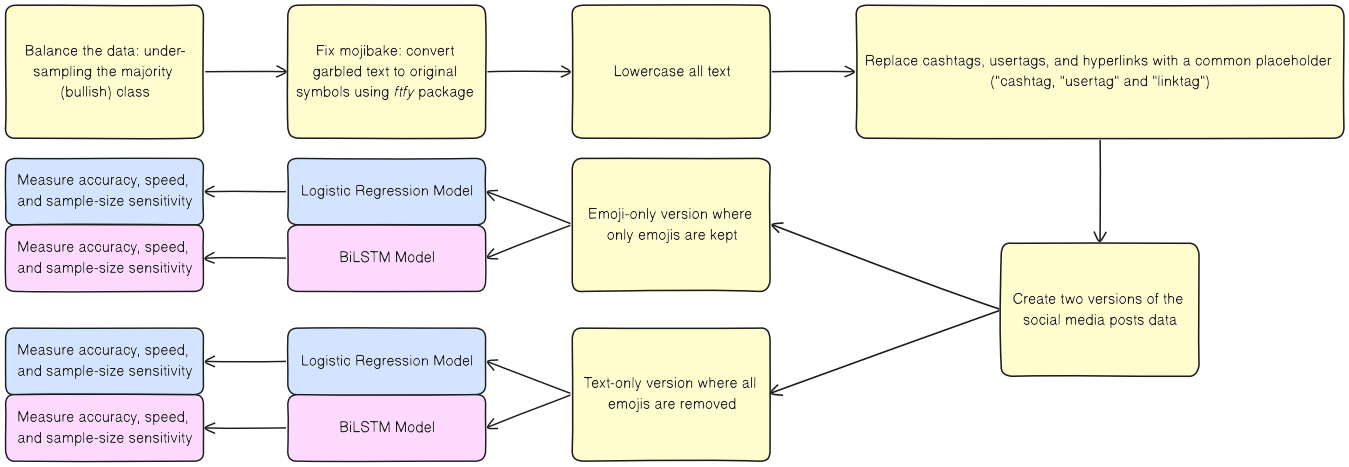

Fig. 4: Our methodology at a glance. Pre-processing (yellow, cf. Sec. V-A) generates two data sets, containing only emojis and only text, respectively. We then feed each data set into both a logistic regression model (blue, cf. Sec. VI-B) and our BiLSTM-based model (pink, cf. Sec. VI-C).

two different words, which they are not.

4. Replacing cashtags, usertags, and hyperlinks: All cashtags (eg: $AAPL, $BTC.X) were replaced by a common word "cashtag". All user mentions (eg: @user1, @elonmusk) were replaced by a common word "usertag". All hyperlinks (eg: https://blog.com/shockingnews) were replaced by a common word "linktag".

5. Creating two versions of the data: After performing the mentioned pre-processing steps, we create two versions of the data: one version where all emojis are removed, and another version where only emojis are retained. Details are provided in Sec. VI.

B. Common but excluded pre-processing Steps

Other pre-processing steps that can be performed include removing stop-words, removing punctuation marks, removing numbers, stemming, lemmatization, and part-of-speech (POS) tagging. However, a seminal contribution by Renault, [21] shows that performing these steps reduces accuracy for various reasons and varying degrees. We have thus omitted the following pre-processing steps otherwise commonly applied to general-purpose natural language processing tasks.

1. Stop-words are a set of commonly used words that are considered insignificant to sentiment analysis, such as the words "a", "the", "is", "are" etc. However, some words that are usually considered stop-words can be important in determining the sentiment of financial posts. Such words include "up", "down", "over", "under", "below", "above" which are all included in the NLTK stop-words corpus. Removing the NLTK stop-words corpus from finance microblogging posts significantly reduces the accuracy of a sentiment analysis model.

2. Removing punctuation marks can also affect the prediction of the sentiment of a financial post. For instance, the post "I can't explain how I'm feeling... :'(" would show a different sentiment compared to "I can't explain how I'm feeling!!! :')". Removing punctuation marks would transform them both into "I cant explain how Im feeling" which would carry much less information about the sentiment compared to the previous examples. Removing punctuation leads to a slight decrease in accuracy.

3. Numbers also carry information that can help determine the sentiment of a post. For example, these two Tweets contain the same characters except for numbers but carry opposite sentiments: "The stock will go from 90 to 150" and "The stock will go from 150 to 90". Therefore, we do not remove numbers.

4. Stemming is a process used to reduce words to their base forms by removing suffixes or prefixes. Stemming slightly reduces the accuracy of a sentiment analysis model in a finance microblogging context. Words like "shorted", "shorts" and "shorties" would all be reduced to the form "short" after stemming. The word "short" is important in a financial context because it is equivalent to "sell". This example shows that stemming might remove important information from a financial post.

5. Part-Of-Speech (POS) tagging involves analyzing each word and giving it a label based on its part of the speech (such as noun, verb, adjective etc.). Again, POS tagging reduces the accuracy.

C. Running Logistic Regression and BiLSTM models on the emoji only data and the text only data

After performing the pre-processing steps and obtaining two versions of the data—emoji-only and text-only, we measure model accuracy, computational efficiency, and

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sample size sensitivity of emoji-only data vs. text-only data using two models: a simple logistic regression model and a more advanced BiLSTM model. The full details of our two models are discussed in the next section.

VI. Implementation Details

In the following, we report the detailed steps followed to obtain our results. The info reported in this section, together with the detailed methodology discussed above, allows for complete reproducibility of our experiments.

A. Pre-processing

We use the Python module "ftfy" to convert mojibake (step *2.* in Sec. V-A). For instance, the following *input*:

> *â€ "> and âœ" werenâ€™t decoded correctly.*
> *Let's use ftfy ðŸŒ to fix that \x85*

can be transformed to the following *output*:

> *—> and ✔ weren't decoded correctly.*
> *Let's use ftfy 🌐 to fix that...*

To generate a "text only" copy of posts containing emojis (step *5.* in Sec. V-A), we replaced any emoji with an empty string. To do so, we use a function from the emoji module: emoji.replace_emoji(). For the "emojis only" copy, we used a function from the emoji module, to extract emojis from text: emoji.emoji_list().

B. Logistic Regression Model

To tokenize the text data, we used a custom tokenization pattern that includes all non-whitespace characters (regular expression `\w+|[^\s]`). Using the default tokenization function of the vectorizer would have eliminated emojis. For vectorization, we used a Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer with default settings from the sklearn library. The data was split into training and testing samples, with the testing sample being 20% of the data. The LogisticRegression() model from the sklearn library was used with the default parameters. The logistic regression model assigns a probability to each row. Rows with a probability of 0.5 or above are labeled as 1.

C. BiLSTM Model

In addition to the logistic regression model, we also utilize a bidirectional LSTM (Long Short-Term Memory) model. The LSTM model is effective in capturing sequential information and dependencies. The bidirectional version of LSTM (BiLSTM) is able to analyze a meaning of a word taking into consideration not just the words before it, but also the words after it.

To enable the model to process textual data, we tokenize the text data using the Twitter-RoBERTa base sentiment tokenizer. This tokenizer, loaded from the `cardiffnlp/twitter-roberta-base-sentiment-latest` model, was trained on over a hundred million Tweets and fine-tuned for sentiment analysis. We set the length of each Tweet to 128 tokens by performing either padding (adding a special padding token) or truncation (removing tokens from the end of the Tweet). We split the tokenized input into training, validation and testing sets, with 15% of the entire data set as testing and 15% of the training data set as validation.

We use a sequential-three layer model starting with an embedding layer, followed by a bidirectional LSTM layer and, finally, a dense layer. The embedding layer converts the tokenized input into dense vector representations with a size of 128, capturing semantic relationships between the tokens. The bidirectional LSTM layer processes the output of the embedding layer in both forward and backward directions. Both the forward and backward LSTM layers have 64 neurons. We set a dropout parameter of 0.2 to prevent overfitting. The dense layer maps the output of the LSTM layer into a single output neuron which is converted into a scalar value using the sigmoid function.

The model is compiled with the binary cross-entropy loss function and the Adam optimizer. The accuracy metric is also specified for evaluation during training. The model is trained using 10 epochs but the training stops if the validation loss does not improve for two consecutive epochs; the model with the lowest validation loss is saved and used. This is done to save time as the training time for each epoch is high due to the large data set size.

VII. Results

In the following, we report and discuss the results obtained by performing the above introduced experiments.

A. Logistic Regression Model

Model performance. Tab. III (top) summarizes the performance metrics using our logistic regression model, whereas the bottom part of the same table summarizes training and prediction times.

As can be seen, emojis-only achieve almost the same performance as text-only sentiment analysis under all metrics while requiring significantly less data.

The training time is the average time taken for training the model 10 times on a sample size of 100,000 posts. The prediction time is the average time taken for predicting the labels of 100,000 posts, which was done 50 times. We observe that using only emojis is not only significantly faster to train (32×), but also faster to predict (almost 2.9×)—the latter being important for high-speed intraday trading.

Sensitivity to data set size. Fig. 5(a) shows the predictive accuracy of using only emojis (left) vs. using only text (right), plotted over data set size. As can be seen in the figure, the model trained on emojis-only data reaches high accuracy much faster, but then degrades, though very mildly. We assume that this is due to significant imbalances in the data, leading the model to assign a fraction of its approximation capacity to memoization. While a full investigation of this effect is left for future work, it underscores that in the case of logistic regression,

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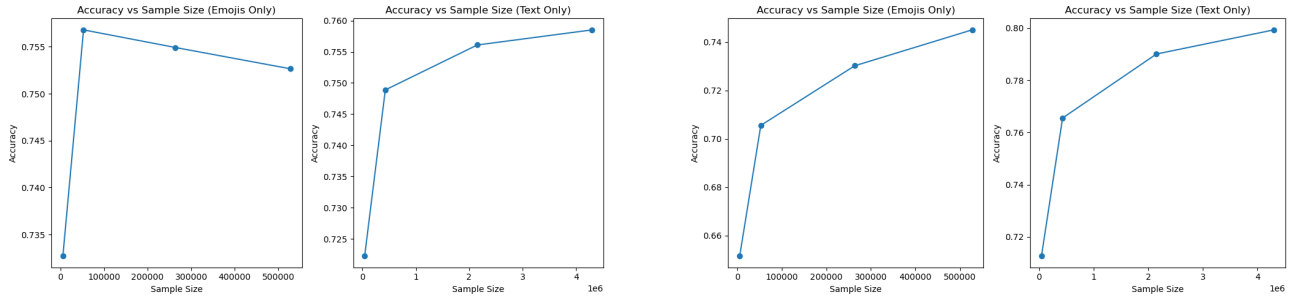

(a) (b)

Fig. 5: Size sensitivity for our (a) logistic regression and (b) LSTM-based models—emojis (left) vs. text (right).

TABLE III: Logistic regression model. Top: accuracy metrics. Bottom: Training and prediction speed.

Performance Metrics	text only	emojis only
Accuracy	0.75	0.75
Recall (weighted avg)	0.75	0.76
Precision (weighted avg)	0.75	0.75
F1 Score (weighted avg)	0.75	0.75
Training Sample Size	3,439,201	422,033

Timings	text only	emojis only
Training time (seconds)	21.10	0.66
Training Sample Size	100,000	100,000
Prediction time (seconds)	0.0021	0.00074
Prediction Sample Size	100,000	100,000

strong performance can be reached using very little data (less than 100,000 posts vs. 4m).

B. BiLSTM Model

Model performance. Like for the logistic regression model, Tab. IV summarizes our results for our BiLSTM-based model. The top half shows our performance metrics, indicating that using only emojis is a good proxy for sentiment that potentially transcends languages. It also demonstrates that significantly heavier machinery (e.g., deep learning) is necessary to harness the additional predictive information contained in (mostly English) text. This boost in performance comes from the sequential nature of text: emoji positions can often be permuted without changing the meaning, resulting in a lack of temporal context for the (Bi-)LSTM. The training time is reported per epoch, as the average of 10 epochs of training the model on 100,000 posts. The prediction time is taken as the average of predicting 100,000 posts 10 times. It is noteworthy that the deep learning approach results in a severely slower prediction speed but boosts text-based performance in terms of accuracy when compared to the faster logistic regression approach by around 5% (absolute). We also note that the times reported for the logistic regression model is the average time *for each complete model fitting*, further increasing the gap in terms of training speed between the two.

TABLE IV: BiLSTM Model. Top: accuracy metrics. Bottom: Training and prediction speed.

Performance Metrics	text only	emojis only
Accuracy	0.80	0.74
Accuracy	0.80	0.74
Recall (weighted avg)	0.80	0.74
Precision (weighted avg)	0.81	0.74
F1 Score (weighted avg)	0.80	0.74
Training Sample Size	3,439,201	422,033
Epochs	6	10

Timings	text only	emojis only
Training time (seconds)	25.04	25.08
Training Sample Size	100,000	100,000
Prediction time (seconds)	7.88	7.85
Prediction Sample Size	100,000	100,000

Sensitivity to data set size. Fig. 5(b) shows the accuracy using only emojis (left) vs. using only text (right). As was the case for logistic regression, using only emojis requires much less data (approx. 500,000 vs. 4m), at the expense of a minor performance gap between the two.

VIII. Conclusion & Future Work

In this contribution, to the best of our knowledge, we are the first in the literature to analyse the possibility of an emojis-only approach for sentiment analysis of financial social media posts. Leveraging a large dataset (18.5M posts) and adopting state of the art techniques (logistic regression and LSTM), we found that, for a given choice of model (logistic regression), an emojis-only approach shows the same accuracy as a text-only approach. Moreover, when the two approaches are compared, the emojis-only approach emerges as the champion: it is 32 times more efficient in training while requiring 40 times less data, and it is almost 3 times more efficient in decision making. The speed advantage is particularly valuable for intra-day high-speed analysis, especially in the volatile cryptocurrency space. In our future work, we wish to extend the setting presented in this paper to include a combined text+emoji approach, as well as to investigate in more details the relationship between training-data and performance.

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